

Total No. of Questions : 7] [Total No. of Printed Pages : 3

Roll No.

REF NO. Conf/Even/N/2025-26/22

**B. Sc. (Hons) Semester IV Examination
2025 - 26 (NEP)**

BOTANY

Paper No. BOTMJ47 : Microbial Resources

Time : 2½ hours

Full Marks : 70

**(Write your Roll No. at the top immediately
on the receipt of this question paper)**

The figures in the right-hand margin indicate marks

**Answer five questions including Question 1
which is compulsory**

**1. Answer the following in less than
100 words each. : 7×2**

(a) What do you understand by probiotics? Explain with a suitable example.

(b) What is the role of mashing in the production of alcoholic beverages?

(c) Define the term caseiculture.

(d) What is chelation? How does it help in phosphate solubilization?

(e) What is gene pyramiding?

(f) What is the role of ATP during nitrogen fixation?

(g) What is a "superbug" in bioremediation?

2. What are antibiotics? Describe their common modes of action with suitable example. How do microorganisms develop resistance to antibiotics? 14

3. What do you understand by biotransformation? Explain the process with suitable examples. 14

4. Explain the role of microorganisms in waste management with special reference to composting of solid wastes, wastewater treatment and methane production pathways. 14

5. Discuss the process of biological nitrogen fixation with special reference to the structure and mechanism of the nitrogenase enzyme complex. 14

(3)

6. Answer any two of the following : 2×7

- (a) Mechanism of action of *Bacillus thuringiensis* as a biopesticide.
- (b) Differentiate between *in situ* and *ex situ* bioremediation.
- (c) Microbial vaccines.

7. Answer any two of the following : 2×7

- (a) Mechanisms of phosphate solubilization by Phosphate solubilizing Bacteria (PSB).
- (b) Explain the role of microorganisms in the production of organic acids; with suitable examples.
- (c) What is mushroom cultivation? Explain in detail the control measures required during mushroom cultivation.

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